Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 Which one of the following statements about 20.3 g of $Co_2(SO_4)_3$ is **incorrect**?
[The molar mass of $Co_2(SO_4)_3$ = 406 g mol^{-1}]

A It contains 1.5×10^{23} ions.

B It contains 0.25 mol of ions.

C It contains 0.15 mol of Co^{3+} ions.

D It contains 47.3 % of oxygen by mass.

2 25.0 cm^3 of a solution of hydrogen peroxide, H_2O_2, was oxidised by excess potassium manganate(VII) solution under acidic condition. The volume of oxygen gas produced at room temperature and pressure was 26 cm^3.

What is the concentration of hydrogen peroxide in the solution?
[The molar volume of a gas at r.t.p. = 24 dm^3 mol^{-1}]

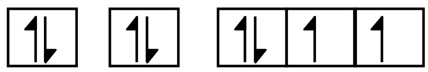
A 2.89×10^{-2} mol dm^{-3}

B 4.33×10^{-2} mol dm^{-3}

C 8.66×10^{-2} mol dm^{-3}

D 1.30×10^{-1} mol dm^{-3}

3 Which one of the following species has the electronic configuration as shown below?



A N

B S

C O^{2-}

D F^{+}

4 The equations for four reactions are given below.

$$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$$
$$2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$$
$$Cl_2 + 2Fe^{2+} \rightarrow 2Cl^- + 2Fe^{3+}$$

What is the correct order of strength of Cl^-, Fe^{2+} and I^- as reducing agents?

	weakest	$\longrightarrow$	***strongest***
A	Cl^-	I^-	Fe^{2+}
B	Cl^-	Fe^{2+}	I^-
C	Fe^{2+}	I^-	Cl^-
D	I^-	Cl^-	Fe^{2+}

5 Which one of the following pairs does the first species have higher ionisation energy than the second?

A Sb, Te **B** Kr, Ar **C** Mg^+, Al^+ **D** Se, Se^+

6 Which one of the following sets of solid substances includes a giant metallic structure, a giant molecular structure and a simple molecular structure?

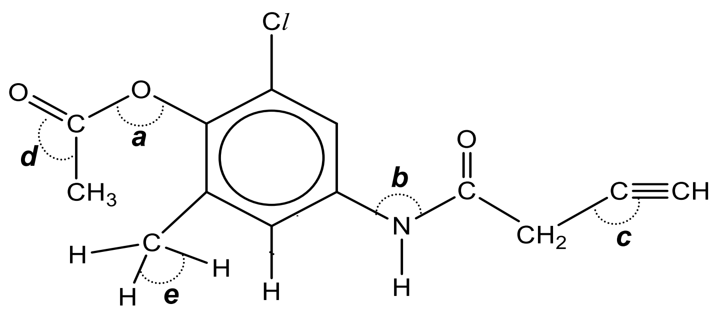
A Al, C, SiO_2

B Pb, Al_2O_3, I_2

C Ni, SiO_2, KI

D Be, Si, SO_3

7 Refer to the molecule below.



Which one of the following shows the correct information with regard to the various labeled bond angles?

	smallest bond angle	⟶	largest bond angle
A	***c***	***d***	***e***
B	***d***	***a***	***c***
C	***e***	***b***	***a***
D	***b***	***e***	***d***

8 Which one of the following pairs of compounds does the first member have a higher boiling point than the second member?

A 2–hydroxybenzoic acid, 4–hydroxybenzoic acid

B ethanol, water

C *trans*–but–2–ene, *cis*–but–2–ene

D butane, 2–methylpropane

9 Which one of the following quantities has the same value as the standard enthalpy change of formation of carbon monoxide?

A ½ $\Delta H_f^\circ(CO_2(g))$

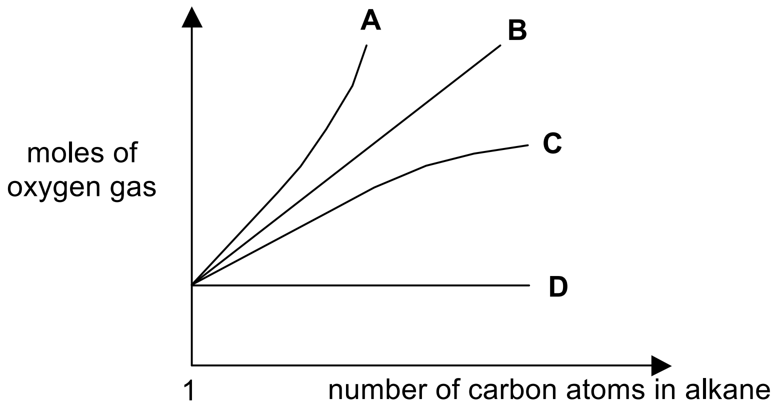
B ½ ΔH_c°(graphite)

C ΔH_c°(graphite) – $\Delta H_c^\circ(CO(g))$

D $\Delta H_f^\circ(CO_2(g))$ – ½ ΔH_c°(graphite)

10 The enthalpy change of combustion of alkanes to produce carbon dioxide and water is an important exothermic reaction.

Which line on the graph shows the relationship between the number of carbon atoms in the alkane and the number of moles of oxygen gas needed for complete combustion of the alkane?



11 The rate of the reaction 3**X**(g) + **Y**(g) → **Z**(g) is given by the rate equation, rate = k [**X**]^2[**Y**].

Two experiments using the same initial amounts of **X** and **Y** were carried out at temperature, *T*. The data is shown below.

experiment	volume of reaction vessel	initial rate of reaction
1	V	R_1
2	$2V$	R_2

Which one of the following shows the relationship between R_1 and R_2?

A $R_2 = \frac{1}{8}R_1$ **B** $R_2 = \frac{1}{4}R_1$ **C** $R_2 = R_1$ **D** $R_2 = 8R_1$

12 The reaction below is zero order with respect to **P** and first order with respect to **Q**.

$$\mathbf{P}(aq) + \mathbf{Q}(aq) \rightarrow \mathbf{R}(aq) + \mathbf{W}(aq)$$

Which one of the following statements is correct?

A **P** is a catalyst in this reaction.

B **Q** is in excess.

C The concentration of **Q** decreases exponentially with time.

D The overall order of reaction is 2.

13 Given that

$$\mathbf{A}_2(g) + 4\mathbf{C}(g) \rightleftharpoons 2\mathbf{AC}_2(g) \qquad K_c = 4.8 \text{ (numerical value)}$$

it follows that, for the reaction,

$$\mathbf{AC}_2(g) \rightleftharpoons \tfrac{1}{2}\,\mathbf{A}_2(g) + 2\mathbf{C}(g) \qquad K_c = \mathbf{Y} \text{ (numerical value)}$$

The value of **Y** would be

A $\dfrac{1}{\sqrt{4.8}}$

B $\dfrac{1}{2.4}$

C $\dfrac{1}{4.8}$

D 2.4

14 The equilibrium constant for the following reaction is less than 1.

$$HPO_4^{2-}(aq) + H_2BO_3^-(aq) \rightleftharpoons H_2PO_4^-(aq) + HBO_3^{2-}(aq)$$

Which one of one the following gives the correct relative strengths of the acids and bases in the reaction?

	acids				bases		
A	$H_2PO_4^-$	>	$H_2BO_3^-$	and	HBO_3^{2-}	>	HPO_4^{2-}
B	$H_2BO_3^-$	>	$H_2PO_4^-$	and	HBO_3^{2-}	>	HPO_4^{2-}
C	$H_2PO_4^-$	>	$H_2BO_3^-$	and	HPO_4^{2-}	>	HBO_3^{2-}
D	$H_2BO_3^-$	>	$H_2PO_4^-$	and	HPO_4^{2-}	>	HBO_3^{2-}

15 What is the pH of the final solution formed by mixing equal volumes of two separate portions of dilute sulfuric acid of pH 2.0 and pH 4.0?

A 2.3 **B** 2.6 **C** 3.0 **D** 3.6

16 The successive ionization energies, in kJ mol^{-1}, of an element **P** are given below:

558 1820 2704 5200

Which group does **P** belong to?

A I

B II

C III

D IV

17 Element **G** is found in Period 3 of the Periodic Table.

The oxide and chloride of element **G** are separately mixed with water. The two resulting solutions have the same effect on litmus paper.

What is the identity of element **G**?

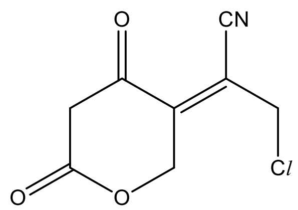
A aluminium

B magnesium

C phosphorus

D sodium

18 Which one of the following options correctly show the correct number of σ and π bonds in the following molecule?



	no. of σ bonds	no. of π bonds
A	13	3
B	13	5
C	19	3
D	19	5

19 Bromine in tetrachloromethane is added separately to benzene, methylbenzene, hexane and hexene in the dark.

Which one of the following pairs will show the same observation?

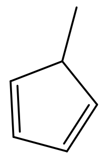
A hexane and benzene

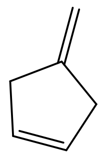
B hexane and hexene

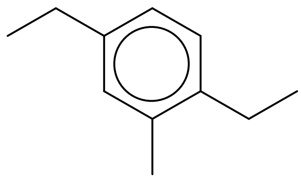
C hexene and benzene

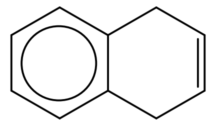
D hexene and methylbenzene

20 Which one of the following will **not** liberate 2 mol of carbon dioxide when 1 mol of the compound is treated with excess hot acidified potassium manganate(VII)?

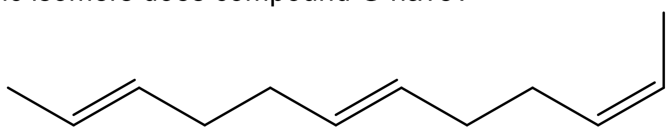
A 

B 

C 

D 

21 How many geometric isomers does compound **G** have?



compound **G**

A 2

B 4

C 6

D 8

22 Compound **H** was subjected to dehydration followed by reaction with acidified potassium manganate(VII). One of the final products formed gives a pale yellow precipitate with alkaline aqueous iodine.

What is the formula for compound **H**?

A $CH_3CH_2CH_2CH_2CH_2OH$

B $CH_3CH(OH)CH_2CH_2CH_3$

C $CH_3CH_2CH(OH)CH_2CH_3$

D $CH_3CH(CH_3)CH(OH)CH_3$

23 In the upper atmosphere, chlorofluoroalkanes (CFCs) are broken down to give chlorine but not fluorine radicals.

What is the best explanation for this?

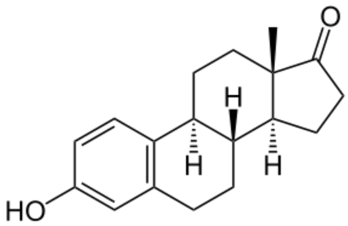
A Fluorine is more electronegative than chlorine.

B Chlorine is more stable than fluorine.

C The C-F bond is stronger than C-C*l* bond.

D The C-F bond is longer than C-C*l* bond.

24 Estrone is one the three estrogenic hormones present in nature and it is the most predominant type of estrogen found in postmenopausal women.



Estrone

Which of the following gives the correct set of observations for estrone?

(i) It gives a pale yellow precipitate on warming with alkaline aqueous iodine.

(ii) It gives an orange precipitate with 2,4-dinitrophenylhydrazine.

(iii) One mole of estrone reacts with sodium to form one mole of hydrogen gas.

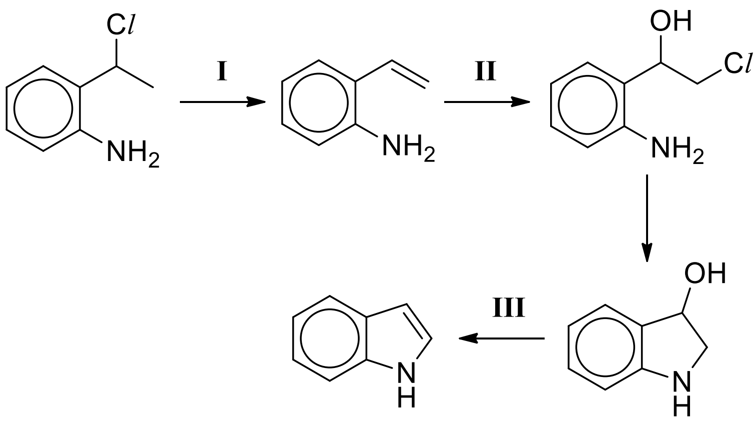
A (ii)

B (i), (ii)

C (ii), (iii)

D (i), (ii), (iii)

25 Indole is an organic compound commonly used in fragrances and the precursor to many pharmaceuticals. One way to synthesize indole is shown below.



Indole

Which one of the following gives the correct reagents required for the steps indicated?

	I	II	III
A	alcoholic KOH	aqueous Cl_2	concentrated H_2SO_4
B	alcoholic KOH	Cl_2 in CCl_4	concentrated H_2SO_4
C	aqueous KOH	aqueous Cl_2	concentrated H_2SO_4
D	aqueous KOH	Cl_2 in CCl_4	aqueous H_2SO_4

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and **3** are correct	**1** and **2** only are correct	**2** and **3** **only** are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 Hydrogen sulfide reacts with water according to the following equilibrium equation.

$$H_2S(aq) + 2H_2O(l) \rightleftharpoons 2H_3O^+(aq) + S^{2-}(aq)$$

Which of the following would cause a shift in equilibrium position when added to the above equilibrium mixture?

1 water

2 sodium carbonate

3 ammonium sulfate

27 Which of the following pairs of substances have the same type of bonding and structure?

1 Rb Kr

2 diamond SiC

3 PH_3 $SiCl_3H$

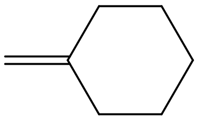
28 Which of the following enthalpy changes are **always** exothermic?

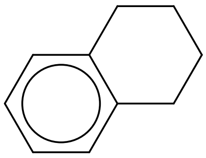
1 Enthalpy change of neutralisation

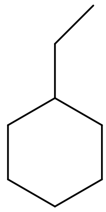
2 Lattice energy

3 Ionisation energy

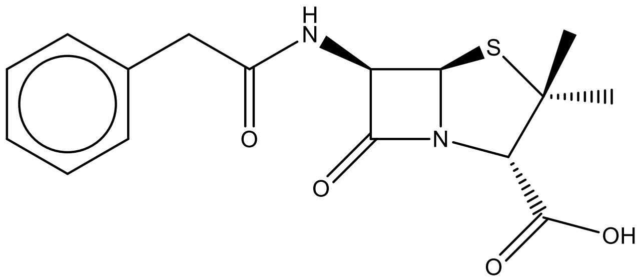
29 Which of the following decolourises a solution of hot acidified potassium manganate(VII)?

1 

2 

3 

30 Benzylpenicillin, commonly known as penicillin G, has the following structure.



Which of the following statements are correct?

1 Penicillin G decolourises acidified potassium manganate(VII).

2 Penicillin G reacts with 2,4-dinitrophenylhydrazine to give an orange precipitate.

3 1 mol of penicillin G reacts with 1 mol of NaOH at room temperature.